

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

*I P.GOWTHAM REDDY(192525139) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: P.GOWTHAM REDDY

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Design pseudocode for a **stateful firewall system** that tracks active connections and filters packets based on session state. Evaluate how state tracking improves security compared to stateless filtering.

ANSWER:

A **stateful firewall** maintains a **state table** containing information about active network connections. Instead of examining each packet independently, it checks whether the packet belongs to a valid, established session.

Pseudocode

START

Create empty STATE_TABLE

For each incoming packet P:

 Extract:

 Source_IP

 Destination_IP

 Source_Port

 Destination_Port

 Protocol

 TCP_Flags

 CONNECTION = identify connection using
 (Source_IP, Destination_IP,
 Source_Port, Destination_Port, Protocol)

 IF CONNECTION exists in STATE_TABLE THEN

 STATE = STATE_TABLE[CONNECTION]

 IF packet matches expected session state THEN

 ALLOW packet

 UPDATE STATE_TABLE

 ELSE

 DROP packet

 LOG "Invalid session packet"

 ELSE

 IF P is a new connection request

 AND firewall policy allows it THEN

 CREATE new entry in STATE_TABLE

 ALLOW packet

```
ELSE
  DROP packet
  LOG "Unauthorized connection"
```

```
END IF
```

```
Remove expired or closed connections
from STATE_TABLE
```

```
END FOR
```

```
STOP
```

Example State Table

Source	Destination	Protocol	State
192.168.1.10:5000	10.0.0.5:80	TCP	ESTABLISHED
192.168.1.15:4500	10.0.0.8:443	TCP	SYN-SENT

For a TCP connection, the firewall can track states such as:

NEW → ESTABLISHED → FIN/TERMINATED

Stateful vs. Stateless Firewall

Feature	Stateless Firewall	Stateful Firewall
Connection tracking	No	Yes
Packet decision	Individual packet	Packet + session state
Security	Lower	Higher
Handles established sessions	Limited	Efficient
Detects unexpected packets	Limited	Better
Resource usage	Lower	Higher
Complexity	Simple	More complex

How State Tracking Improves Security

In **stateless filtering**, each packet is evaluated independently using rules such as source IP, destination IP, port, and protocol. An attacker may therefore send packets that appear to satisfy the basic rules even though they are not part of a legitimate connection.

A **stateful firewall** verifies that packets belong to an existing and valid session. For example, if an internal computer establishes a legitimate TCP connection to a web server, response packets from that server can be allowed because they match the recorded session. An unexpected packet claiming to belong to that session can be rejected.

Therefore, state tracking improves security by:

1. **Preventing unauthorized packets** that do not belong to an active session.
2. **Detecting invalid connection states**, such as unexpected TCP packets.
3. **Allowing legitimate return traffic** without requiring separate rules for every response.
4. **Reducing exposure to spoofed or unsolicited traffic.**
5. **Providing better session-level visibility** for logging and monitoring.

Conclusion

A stateful firewall provides stronger protection than stateless filtering because it maintains **context about active connections**. Although it requires more memory and processing to maintain the state table, it offers significantly better control over legitimate and unauthorized network sessions.

2. Design pseudocode for a **signature-based malware detection system** that scans files and compares them against a known signature database. Evaluate its effectiveness in detecting known threats and its limitations against new attacks.

ANSWER:

A **signature-based malware detection system** identifies malicious files by comparing their characteristics, such as hashes or byte patterns, with a database of **known malware signatures**.

Pseudocode

START

Load KNOWN_SIGNATURE_DATABASE

For each FILE in target directory:

 Calculate FILE_HASH

 Extract FILE_SIGNATURE

 IF FILE_HASH matches KNOWN_SIGNATURE_DATABASE THEN

 Mark FILE as "MALWARE"

 Block or quarantine FILE

 Log detection

 CONTINUE

 ELSE IF FILE_SIGNATURE matches a known malware pattern THEN

 Mark FILE as "MALWARE"

 Block or quarantine FILE

 Log detection

 CONTINUE

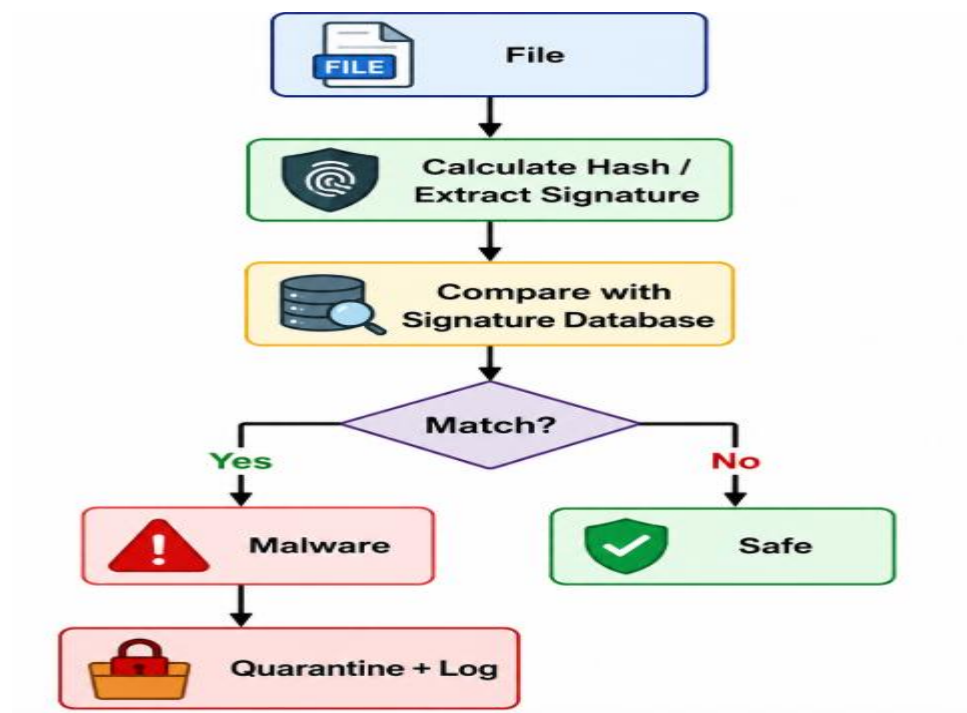
```
ELSE
  Mark FILE as "SAFE"
END IF
```

```
END FOR
```

```
Display detection results
```

```
STOP
```

Working Process



Example

Suppose the signature database contains:

Malware Signature/Hash

Virus_A A12345

Trojan_B B67890

Worm_C C54321

If a scanned file produces hash B67890, the system identifies it as **Trojan_B** and quarantines the file.

Effectiveness

Advantages:

- **Highly effective against known malware:** Previously identified malware can be detected accurately.
- **Fast scanning:** Hash and signature comparisons can be performed efficiently.
- **Low false positives:** Exact signatures provide strong evidence of a known threat.
- **Simple to implement:** The detection process is straightforward.
- **Easy to update:** New malware signatures can continuously be added to the database.

Limitations Against New Attacks

The major weakness is that signature-based detection depends on **known signatures**.

Threat	Detection
Known malware	✓ Very effective
Previously identified variants	✓ Usually effective
Completely new malware	✗ May be missed
Zero-day attacks	✗ Often missed
Polymorphic malware	⚠ May evade fixed signatures
Obfuscated malware	⚠ Difficult to detect

For example, if an attacker creates a new malware program whose signature does not exist in the database, the system may classify it as safe even though it is malicious.

Conclusion

A **signature-based malware detection system** is highly effective, fast, and reliable for detecting **known threats**, but it has limited ability to detect **zero-day, polymorphic, and previously unseen malware**. Therefore, modern security systems often combine signature detection with **behavior-based analysis, heuristic detection, and machine learning** to improve protection against new attacks.

3. Design pseudocode for a **distributed denial-of-service (DDoS) detection system** using traffic threshold analysis. Evaluate how the algorithm differentiates between legitimate high traffic and attack traffic.

ANSWER:

1. Objective

The system monitors incoming network traffic and detects a possible **Distributed Denial-of-Service (DDoS) attack** when traffic exceeds predefined thresholds for a sustained period.

2. Pseudocode

BEGIN

Set NORMAL_THRESHOLD
Set CRITICAL_THRESHOLD
Set TIME_WINDOW
Set SUSTAINED_COUNT = 0

WHILE network is running

 Collect incoming traffic during TIME_WINDOW

 Calculate:

 Traffic_Rate
 Number_of_Source_IPs
 Request_Rate
 Average_Packet_Size

 IF Traffic_Rate <= NORMAL_THRESHOLD THEN
 Status = "Normal Traffic"
 SUSTAINED_COUNT = 0

 ELSE IF Traffic_Rate > NORMAL_THRESHOLD
 AND Traffic_Rate <= CRITICAL_THRESHOLD THEN

 Check traffic characteristics

 IF Source_IPs are diverse
 AND requests are valid
 AND traffic pattern is consistent THEN

 Status = "Legitimate High Traffic"
 SUSTAINED_COUNT = 0

 ELSE
 SUSTAINED_COUNT = SUSTAINED_COUNT + 1
 END IF

 ELSE IF Traffic_Rate > CRITICAL_THRESHOLD THEN

 SUSTAINED_COUNT = SUSTAINED_COUNT + 1

 IF Source_IPs are highly distributed
 AND request rate is abnormally high
 AND traffic continues for multiple TIME_WINDOWs THEN

 Status = "DDoS Attack Detected"
 Trigger Alert
 Apply Traffic Filtering
 Block/Scrub Malicious Traffic

Log Attack Details

ELSE

Status = "Possible Legitimate Traffic"

END IF

END IF

END WHILE

END

3. How the Algorithm Differentiates Traffic

Feature	Legitimate High Traffic	DDoS Attack Traffic
Traffic volume	High but expected	Extremely high/unexpected
Duration	Usually temporary or predictable	Often sustained
Source IPs	Diverse and legitimate	Large number of distributed sources
Requests	Valid and meaningful	Repetitive, abnormal, or malformed
Traffic pattern	Related to genuine user activity	Sudden and unusual
Response	Allow/monitor	Alert, filter, or block

4. Example

Suppose a web server normally receives **1,000 requests/sec**.

- Normal threshold = **2,000 requests/sec**
- Critical threshold = **5,000 requests/sec**

If traffic rises to **4,000 requests/sec** because of a legitimate product launch, the system checks the source IPs and request patterns. If the requests are valid and users are behaving normally, it classifies the event as **legitimate high traffic**.

If traffic suddenly reaches **10,000 requests/sec**, comes from thousands of distributed sources, contains repetitive requests, and remains high across several time windows, the system identifies it as a **DDoS attack**.

5. Evaluation

Advantages:

- Simple and fast to implement.
- Requires relatively low computational resources.
- Can detect sudden traffic spikes quickly.
- Thresholds can be adjusted according to normal network behavior.

Limitations:

- A fixed threshold may generate **false positives** during legitimate events such as flash crowds.
- Low-rate or slowly increasing DDoS attacks may remain below the threshold.
- Threshold analysis alone cannot always determine whether traffic is malicious.

Conclusion:

Traffic threshold analysis is effective as a **first-level DDoS detection mechanism**, especially for sudden high-volume attacks. To improve accuracy, it should combine traffic thresholds with **source-IP analysis, request behavior, historical traffic baselines, and anomaly detection**.

4. Design pseudocode for a **network intrusion detection system using machine learning classification**. Evaluate how feature selection and training data impact detection accuracy.

ANSWER:

1. Objective

The system monitors network traffic, extracts important features, and uses a **machine learning classifier** to determine whether the traffic is **Normal** or an **Intrusion/Attack**.

2. Pseudocode

BEGIN

Load Network Traffic Dataset

Clean Dataset

Remove missing or duplicate records

Normalize numerical features

Select Important Features

Remove irrelevant features

Calculate feature importance

Keep high-value features

Split Dataset into:

Training Data

Testing Data

Train Machine Learning Classifier

Input: Selected Features + Training Labels

Output: Trained Intrusion Detection Model

FOR each incoming network packet/connection

Extract network features

Select the same important features

Preprocess features

Prediction = Model.predict(features)

IF Prediction == "Normal" THEN

 Allow Traffic

 Log Normal Activity

ELSE IF Prediction == "Attack" THEN

 Generate Intrusion Alert

 Block or Isolate Traffic

 Log Attack Details

END IF

END FOR

Evaluate Model using Testing Data

Calculate:

 Accuracy

 Precision

 Recall

 F1-Score

END

3. Basic Flow

Network Traffic



Feature Extraction



Feature Selection



Data Preprocessing



Training Dataset



ML Classification Model



Prediction



Normal



Attack



Allow



Alert/Block

4. Impact of Feature Selection

Feature selection determines which network characteristics are given to the machine-learning model.

Examples of useful features:

- Source and destination IP
- Source and destination port
- Protocol
- Packet size
- Connection duration
- Number of packets
- Number of requests

Good feature selection:

- Removes irrelevant information.
- Reduces model complexity.
- Decreases training time.
- Can improve detection accuracy.
- Helps reduce false positives.

Poor feature selection:

- Important attack characteristics may be removed.
- Irrelevant features can confuse the classifier.
- Detection accuracy may decrease.
- Training can become slower.

5. Impact of Training Data

The quality and diversity of training data strongly affect the IDS.

Training Data	Effect
Large and diverse dataset	Better generalization
Balanced normal/attack samples	Better detection of different classes
High-quality labeled data	More reliable predictions
Outdated attack samples	Poor detection of new attacks
Imbalanced dataset	Model may favor the majority class
Noisy/mislabeled data	Reduced accuracy

6. Evaluation

The system should be evaluated using:

Accuracy

Accuracy = Correct Predictions / Total Predictions

Precision

Precision = True Positives / (True Positives + False Positives)

Recall

Recall = True Positives / (True Positives + False Negatives)

F1-Score

F1 = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

7. Conclusion

A machine-learning-based NIDS can identify both normal and malicious network traffic automatically. **Feature selection improves efficiency and can increase accuracy by focusing on relevant network characteristics**, while **high-quality, diverse, balanced training data improves the model's ability to detect different types of attacks**. Therefore, both feature selection and training-data quality are critical for achieving a reliable intrusion detection system.

5. Design pseudocode for a secure network access control system that authenticates users and enforces role-based access policies. Evaluate how access control mechanisms enhance overall network security.

ANSWER:

1. Objective

The system authenticates users before allowing network access and then applies **Role-Based Access Control (RBAC)** to determine which network resources the user can access.

2. Pseudocode

BEGIN

Initialize User_Database
Initialize Role_Permissions
Initialize Access_Log

User requests network access

INPUT Username
INPUT Password

```
IF Username exists in User_Database THEN

    Verify Password

    IF Password is correct THEN

        Perform Multi-Factor Authentication (MFA)

        IF MFA is successful THEN

            Retrieve User_Role

            Check Role_Permissions for User_Role

            Display Available Network Resources

            User requests a Resource

            IF Resource is permitted for User_Role THEN

                Grant Access
                Log "Access Granted"

            ELSE

                Deny Access
                Log "Access Denied"
                Generate Security Alert

            END IF

        ELSE

            Deny Access
            Log "MFA Failed"
            Generate Security Alert

        END IF

    ELSE

        Deny Access
        Log "Invalid Password"

        IF Failed_Attempts > Maximum_Attempts THEN
            Lock User Account
            Generate Security Alert
        END IF

    END IF

END IF
```

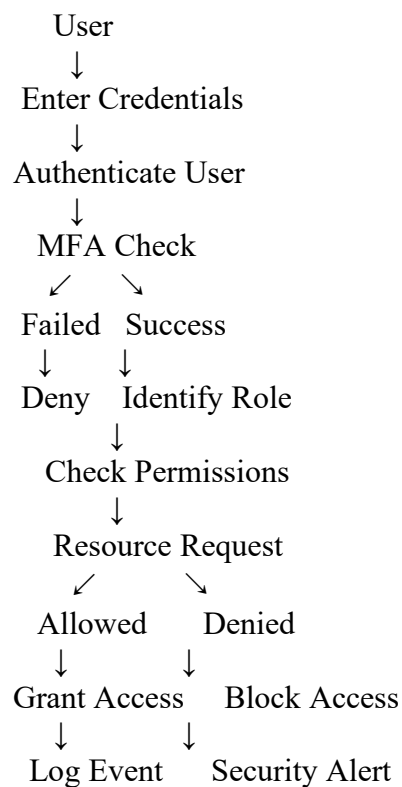
ELSE

Deny Access
Log "Unknown User"
Generate Security Alert

END IF

END

3. System Flow



4. Example of Role-Based Permissions

Role	Network Access
Administrator	Full network and system access
Manager	Business applications and reports
Employee	Approved applications and shared resources
Guest	Limited internet/guest resources

For example, if an **Employee** tries to access an administrator-only server, the system checks the employee's role and **denies the request**.

5. How Access Control Enhances Network Security

1. Prevents unauthorized access

Authentication ensures that only verified users can enter the network.

2. Limits privileges

RBAC follows the **principle of least privilege**, giving users only the permissions required for their jobs.

3. Reduces insider threats

Even an authenticated employee cannot automatically access every network resource.

4. Protects sensitive resources

Critical servers, databases, and administrative systems can be restricted to authorized roles.

5. Improves accountability

Access logs record successful and failed access attempts, helping administrators investigate suspicious activity.

6. Reduces attack impact

If an account is compromised, role-based restrictions limit what the attacker can access.

6. Evaluation

Mechanism	Security Benefit
Username & Password	Basic identity verification
MFA	Provides an additional authentication layer
RBAC	Restricts resources according to user roles
Least Privilege	Minimizes unnecessary permissions
Account Lockout	Prevents repeated login attacks
Access Logging	Provides monitoring and accountability
Security Alerts	Helps detect suspicious activity

7. Conclusion

A secure network access control system combines **strong authentication, MFA, RBAC, least privilege, and continuous logging** to protect network resources. Authentication verifies **who the user is**, while RBAC determines **what the user is allowed to access**. These mechanisms significantly reduce unauthorized access, insider threats, and the potential damage caused by compromised accounts.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient